

Assignment #01

Name: Fateh Ali

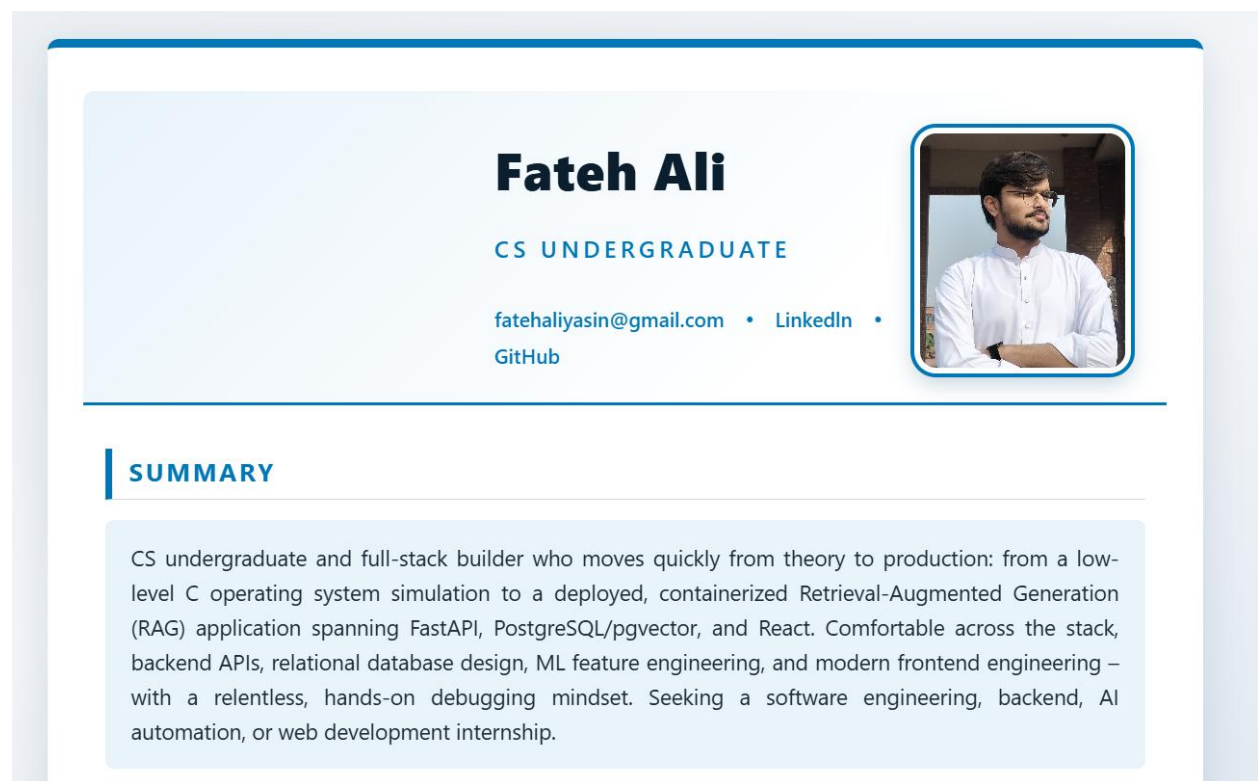
Reg. No: FA24-BCS-032

Assignment 1 — Your Personal CV Page

Difficulty: Foundational · **Covers:** Lectures 3–6 (HTML & HTML5, Semantic HTML and Forms, CSS Fundamentals, the CSS Box Model) · **Folder:** assignment-1/

Task: Build a single-page personal CV/résumé as a static HTML + CSS website — no JavaScript, no framework, just semantic HTML5 and hand-written CSS. This is where you prove you actually know the raw building blocks before anything is allowed to hide them from you.

Screenshots:



EDUCATION

BS Computer Science

COMSATS University Islamabad (Lahore Campus)

09/2024 – Present

TECHNICAL SKILLS

Languages & Concepts: Python, JavaScript, C++, SQL, Java, Object-Oriented Programming (OOP), Relational Database Design, RESTful APIs, ETL Pipelines, Web Scraping, Feature Engineering

Frameworks & Tools: FastAPI, React, Vite, Tailwind CSS, Docker, PostgreSQL, pgvector, SQLAlchemy 2.0, Alembic, JWT Authentication, Git/GitHub, Typer, CustomTkinter, BeautifulSoup4, Loguru, SSMS, Microsoft Office

CERTIFICATES

- **Python for Everybody Specialization** - University of Michigan (Coursera)
- **Python Crash Course** - Google (Coursera)
- **Technical Volunteer, ACM Student Chapter** - Organized and facilitated technical events and academic workshops

PROJECTS

RepoMind– Full-Stack RAG Application

Python, FastAPI, React

- Engineered and deployed a production-ready Retrieval-Augmented Generation (RAG) application from scratch, containerizing a FastAPI backend with a PostgreSQL + pgvector database using Docker, and managing schema evolution via SQLAlchemy 2.0 ORM and Alembic migrations.
- Built a custom document ingestion pipeline that chunks text with overlapping boundaries, and integrated the Google Gemini Embeddings API to perform cosine-similarity search and generate hallucination-free answers via Groq's Llama-3.1-8b-instant model.
- Built a responsive React + Vite + Tailwind CSS frontend with JWT-based session persistence and a Markdown-rendered chat interface; deployed live on Vercel (frontend) and Railway (backend).

Tech Pulse Data Pipeline

Python, BeautifulSoup, Typer

- Engineered an automated pipeline utilizing requests and BeautifulSoup4 to extract, filter, and score unstructured web data based on weighted keyword density.
- Architected a modular CLI with Typer and Loguru to orchestrate data transformation, secure CSV storage, and automated HTML email dispatch via SMTP.

AutoFile Pro

Python, CustomTkinter

- Built a desktop file-management application with a modern dark-mode GUI, featuring intelligent file organization by extension, size, or modification age.
- Implemented threaded backup execution with optional zip compression to keep the UI responsive during large file operations, and a configurable cleanup tool with safe delete-to-recycle-bin/archive handling.

Code:

Index.html:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Resume</title>
    <link rel="stylesheet" href="styles.css" />
  </head>

  <body>
    <div id="main" class="resume">
      <header class="header">
        <div class="header-content">
          <h1>Fateh Ali</h1>
          <p>CS Undergraduate</p>
          <div class="urls">
            <ul id="creds">
              <li>
```

```

links"
    <a href="mailto:fatehaliyasin@gmail.com" class="creds-
links"
        >fatehaliyasin@gmail.com</a
    >
</li>
<li>
    <a
        href="https://linkedin.com/in/fateh-ali-072348352"
        class="creds-links"
        target="_blank"
    >LinkedIn</a
    >
</li>
<li>
    <a
        href="https://github.com/FatehAli02"
        class="creds-links"
        target="_blank"
    >GitHub</a
    >
</li>
</ul>
</div>
<div class="img">
    
</div>
</div>
</header>

```

<section>

<h2>Summary</h2>

<p>

CS undergraduate and full-stack builder who moves quickly from theory to production: from a low-level C operating system simulation to a deployed, containerized Retrieval-Augmented Generation (RAG) application spanning FastAPI, PostgreSQL/pgvector, and React. Comfortable across the stack, backend APIs, relational database design, ML feature engineering, and modern frontend engineering - with a relentless, hands-on debugging mindset. Seeking a software

engineering, backend, AI automation, or web development internship.

</p>
</section>

<section>
 <h2>Education</h2>
 <h4>BS Computer Science</h4>
 <p>COMSATS University Islamabad (Lahore Campus)</p>
 <p>09/2024 – Present</p>
</section>

<section>
 <h2>Technical Skills</h2>
 <p>
 Languages & Concepts: Python, JavaScript, C++,
SQL,
 Java, Object-Oriented Programming (OOP), Relational Database
Design,
 RESTful APIs, ETL Pipelines, Web Scraping, Feature Engineering
</p>
 <p>
 Frameworks & Tools: FastAPI, React, Vite,
Tailwind
 CSS, Docker, PostgreSQL, pgvector, SQLAlchemy 2.0, Alembic, JWT
Authenticat ion, Git/GitHub, Typer, CustomTkinter,
BeautifulSoup4,
 Loguru, SSMS, Microsoft Office
</p>
</section>

<section>
 <h2>Certificates</h2>

 Python for Everybody Specialization - University of
Michigan
 (Coursera)

 Python Crash Course - Google (Coursera)

 Technical Volunteer, ACM Student Chapter - Organized
and

```
        facilitated technical events and academic workshops
    </li>
</ul>
</section>
```

```
<section>
  <h2>Projects</h2>
  <article class="project">
    <h3>RepoMind- Full-Stack RAG Application</h3>
    <p class="technologies"><i>Python, FastAPI, React</i></p>
    <ul>
```

Augmented

```
      <li>
        Engineered and deployed a production-ready Retrieval-
```

and

```
        Generation (RAG) application from scratch, containerizing a
        FastAPI backend with a PostgreSQL + pgvector database using
        Docker, and managing schema evolution via SQLAlchemy 2.0 ORM
```

with

```
        Alembic migrations.
      </li>
      <li>
```

generate

```
        Built a custom document ingestion pipeline that chunks text
```

model.

```
        overlapping boundaries, and integrated the Google Gemini
        Embeddings API to perform cosine-similarity search and
```

```
        hallucination-free answers via Groq's llama-3.1-8b-instant
```

```
      </li>
      <li>
```

```
        Built a responsive React + Vite + Tailwind CSS frontend with
        JWT-based session persistence and a Markdown-rendered chat
        interface; deployed live on Vercel (frontend) and Railway
        (backend).
      </li>
    </ul>
  </article>
```

```
<article class="project">
  <h3>Tech Pulse Data Pipeline</h3>
  <p class="technologies"><i>Python, BeautifulSoup, Typer</i></p>
  <ul>
    <li>
```

```

        Engineered an automated pipeline utilizing requests and
        BeautifulSoup4 to extract, filter, and score unstructured
web data
        based on weighted keyword density.
    </li>
    <li>
        Architected a modular CLI with Typer and Loguru to
orchestrate
        data transformation, secure CSV storage, and automated HTML
email
        dispatch via SMTP.
    </li>
</ul>
</article>

<article class="project">
    <h3>AutoFile Pro</h3>
    <p class="technologies"><i>Python, CustomTkinter</i></p>
    <ul>
        <li>
            Built a desktop file-management application with a modern
            dark-mode GUI, featuring intelligent file organization by
            extension, size, or modification age.
        </li>
        <li>
            Implemented threaded backup execution with optional zip
            compression to keep the UI responsive during large file
            operations, and a configurable cleanup tool with safe
            delete-to-recycle-bin/archive handling.
        </li>
    </ul>
</article>
</section>
</div>
</body>
</html>

```

Styles.css:

```

:root {
    --accent: #0077b6;
    --accent-dark: #005a8c;

```

```
--accent-soft: #e8f3fa;
--text: #222222;
--text-muted: #555555;
--border: #d9d9d9;
}

* {
  box-sizing: border-box;
}

html {
  scroll-behavior: smooth;
}

::selection {
  background: var(--accent);
  color: white;
}

body {
  font-family: "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
  background: linear-gradient(160deg, #e9eef3 0%, #f5f7fa 50%, #dfe8f0
100%);
  background-attachment: fixed;
  color: var(--text);
  line-height: 1.6;
  padding: 40px 20px;
  margin: 0;
  -webkit-font-smoothing: antialiased;
}

.resume {
  width: 100%;
  max-width: 800px;
  margin: 30px auto;
  background-color: white;
  padding: 40px;
  border-top: 6px solid var(--accent);
  border-radius: 8px;
  box-shadow: 0 10px 35px rgba(0, 40, 80, 0.15);
  animation: fadeUp 0.6s ease both;
}
```

```
@keyframes fadeUp {
  from {
    opacity: 0;
    transform: translateY(15px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

.header {
  background: linear-gradient(135deg, var(--accent-soft), #ffffff 70%);
  border-bottom: 2px solid var(--accent);
  border-radius: 8px 8px 0 0;
  padding: 25px 25px 20px;
  margin: -10px -15px 30px;
}

.header-content {
  display: grid;
  grid-template-columns: 1fr 1.9fr auto;
  grid-template-rows: auto;
  row-gap: 5px;
}

.header-content h1 {
  grid-row: 1;
  grid-column: 2;
  margin: 0;
  font-size: 36px;
  font-weight: 800;
  letter-spacing: 0.5px;
  color: #0b1f2e;
  padding-left: 70px;
}

.header-content p {
  grid-row: 2;
  grid-column: 2;
  margin: 5px 0 10px;
  color: var(--accent);
  font-size: 16px;
}
```

```
font-weight: 600;
text-transform: uppercase;
letter-spacing: 3px;
padding-left: 70px;
}

.header-content .urls {
  grid-row: 3;
  grid-column: 2;
  align-self: end;
  padding-left: 70px;
}

#creds {
  list-style: none;
  padding: 0;
  margin: 0;
}

.header-content #creds li {
  display: inline;
}

.header-content #creds li + li::before {
  content: "•";
  color: var(--accent);
  margin: 0 10px;
}

.header-content .img {
  grid-column: 3;
  grid-row: 1 / span 3;
  display: flex;
  align-items: center;
  justify-content: flex-end;
}

.Profile_Pic {
  width: 150px;
  height: 170px;
  object-fit: cover;
  border-radius: 15px;
  border: 4px solid white;
}
```

```
outline: 3px solid var(--accent);
box-shadow: 0 6px 16px rgba(0, 60, 100, 0.25);
transition: transform 0.3s ease;
}

.Profile_Pic:hover {
  transform: scale(1.04) rotate(-1deg);
}

.creds-links {
  color: var(--accent);
  text-decoration: none;
  font-size: 14px;
  font-weight: 600;
  padding-bottom: 1px;
  border-bottom: 2px solid transparent;
  transition:
    color 0.2s ease,
    border-color 0.2s ease;
}

.creds-links:hover,
.creds-links:focus-visible {
  color: var(--accent-dark);
  border-bottom-color: var(--accent);
  outline: none;
}

section {
  margin-bottom: 28px;
}

section h2 {
  margin-bottom: 14px;
  padding: 2px 0 6px 12px;
  border-left: 5px solid var(--accent);
  border-bottom: 1px solid var(--border);
  color: var(--accent);
  font-size: 17px;
  font-weight: 700;
  letter-spacing: 2px;
  text-transform: uppercase;
}
```

```
section p {
  margin: 6px 0;
}

section:nth-of-type(1) p {
  background: var(--accent-soft);
  border-radius: 6px;
  padding: 14px 18px;
  font-size: 15px;
  color: #1c2b36;
  text-align: justify;
}

section h4 {
  margin: 8px 0 3px;
  font-size: 16px;
  color: #0b1f2e;
}

section ul {
  padding-left: 20px;
}

section li {
  margin-bottom: 7px;
}

section li::marker {
  color: var(--accent);
}

section strong,
section b {
  color: #0b1f2e;
}

section p strong {
  color: var(--accent);
}

.project {
  margin-bottom: 20px;
}
```

```
padding: 12px 16px;
border-left: 3px solid #cfe3f0;
border-radius: 0 6px 6px 0;
transition:
  background 0.25s ease,
  border-color 0.25s ease,
  transform 0.25s ease;
}

.project:hover {
  background: #f7fbfd;
  border-left-color: var(--accent);
  transform: translateX(3px);
}

.project h3 {
  margin: 0 0 6px;
  font-size: 17px;
  color: #0b1f2e;
}

.technologies {
  margin: 0 0 8px;
  font-size: 13px;
}

.technologies i {
  display: inline-block;
  font-style: normal;
  font-weight: 600;
  color: var(--accent-dark);
  background: var(--accent-soft);
  border: 1px solid #cfe3f0;
  border-radius: 20px;
  padding: 2px 12px;
}

.project ul {
  margin-top: 5px;
}

.project li {
  line-height: 1.55;
```

```
    font-size: 14.5px;
}

::-webkit-scrollbar {
    width: 10px;
}

::-webkit-scrollbar-track {
    background: #e9eef3;
}

::-webkit-scrollbar-thumb {
    background: var(--accent);
    border-radius: 10px;
}

@media (max-width: 700px) {
    body {
        padding: 15px 10px;
    }

    .resume {
        padding: 25px 20px;
        margin: 10px auto;
    }

    .header {
        margin: -5px -5px 25px;
        padding: 20px 15px;
    }

    .header-content {
        grid-template-columns: 1fr;
        justify-items: center;
        text-align: center;
    }

    .header-content .img {
        grid-column: 1;
        grid-row: 1;
        justify-content: center;
        margin-bottom: 12px;
    }

    .header-content h1,
```

```
.header-content p,  
.header-content .urls {  
  grid-column: 1;  
  padding-left: 0;  
}  
  
.header-content h1 {  
  grid-row: 2;  
  font-size: 28px;  
}  
  
.header-content p {  
  grid-row: 3;  
  letter-spacing: 2px;  
}  
  
.header-content .urls {  
  grid-row: 4;  
}  
  
section:nth-of-type(1) p {  
  text-align: left;  
}  
}  
  
@media print {  
  body {  
    background: white;  
    padding: 0;  
  }  
  
  .resume {  
    box-shadow: none;  
    border-radius: 0;  
    margin: 0;  
    max-width: 100%;  
    animation: none;  
  }  
  
  .header {  
    background: none;  
  }  
  
  .project {  
    break-inside: avoid;  
  }  
}
```

```
transform: none !important;
background: none !important;
}

.creds-links {
  color: var(--text);
}
}
```